ANY CONCEPCION GONZALEZ RODRIGUES

Molecular Bioengineer | Neural Electrophysiology, HD-MEA & Cell Culture

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RESEARCH PROFILE

Molecular Bioengineer with hands-on experience in neuronal cell culture, microelectrode array electrophysiology and experimental workflow development. Established and optimized *ex vivo* retinal HD-MEA recordings at CRTD and conducted functional studies of human and rat olfactory cultures at DZNE. Particularly interested in combining electrophysiology with human brain organoid models to investigate neural plasticity, learning and biological computation.

EDUCATION

MSc in Molecular Bioengineering | University of Porto - ICBAS/FEUP
Sep 2022 - Oct 2024 | Porto, Portugal

Thesis: Establishing CMOS-MEA Recordings in Adult Zebrafish Retina

Supervisor: Prof. Dr. Michael Brand - CRTD, Germany

BSc in Bioengineering | University of Porto - ICBAS/FEUP
Sep 2018 - Jun 2022 | Porto, Portugal

Thesis: Development of Optical Sensors Using Molecularly Imprinted Polymers for Ferritin Detection

Supervisor: Dr. Margarida Piloto - ISEP, Portugal

RESEARCH EXPERIENCE

Master's Thesis Researcher – Neural Electrophysiology
Center for Regenerative Therapies Dresden (CRTD), TU Dresden | Feb 2024 – Aug 2024

- Established and optimized an *ex vivo* HD-MEA workflow for adult zebrafish retina, combining dissection, perfusion, light stimulation and pharmacological experiments.
- Optimized tissue attachment and recording conditions, troubleshooting experimental and analytical challenges.
- Processed HD-MEA recordings using 3Brain BrainWave, configuring analysis parameters and exporting neuronal activity data for further statistical analysis and visualization in R and Python.

Research Intern – Neural Cell Culture and Co-culture
German Center for Neurodegenerative Diseases (DZNE), Dresden | Sep 2023 – Feb 2024

- Seeded and maintained human and rat olfactory cell cultures in 6-well and 24-well plates, including regular medium changes and culture monitoring.
- Prepared olfactory bulb–olfactory epithelium co-cultures for functional studies.
- Prepared media and experimental solutions, following established culture protocols.
- Performed Axion Biosystems MEA recordings during odorant and neurotransmitter stimulation, analyzing neuronal firing and bursting activity.
- Documented experiments and discussed results within an international neuroscience research team.

Research Intern
BioMark, ISEP, Porto | Feb 2022 – Jul 2022

- Contributed to optical sensor development for ferritin detection using molecularly imprinted polymers and UV–Vis spectroscopy

LABORATORY & TECHNICAL SKILLS

Cell culture and experimental biology: Aseptic culture techniques; olfactory cell cultures and co-cultures; culture maintenance; media and solution preparation; biological sample handling.

Molecular biology and imaging: DNA extraction; conventional PCR; agarose gel electrophoresis and gel imaging; immunofluorescence; confocal microscopy.

Electrophysiology and tissue preparation: MEA and HD-MEA recordings (Axion BioSystems, 3Brain); adult zebrafish retinal dissection; *ex vivo* tissue preparation; perfusion; visual and pharmacological stimulation; protocol optimization.

Data analysis: Python and R for neuronal activity analysis, statistical comparisons and visualization; spike detection and sorting using 3Brain BrainWave; firing-rate and stimulus-response analysis.

Research practices: Experimental documentation, troubleshooting, scientific figures and presentations, collaboration in international research teams.

PROFESSIONAL EXPERIENCE

Account Management Intern
MaxWell Biosystems, Zurich | May 2025 – Apr 2026

- Supported scientific communication on HD-MEA applications involving neural cultures, organoids and tissue preparations.
- Worked with scientific and commercial teams to organize technical information and support webinars and research-focused communications.

TRAINING AND CERTIFICATIONS

- FELASA-accredited laboratory animal training LAS modules C01 and F09-F11
- Zebrafish handling course
- Introduction to Python and Machine Learning for the Biosciences (i3S)

LANGUAGES

Portuguese — native (C2); English — C1; Spanish — B1; German — A1

RESEARCH INTERESTS

Organoid intelligence, neural plasticity, closed-loop electrophysiology and biological computing.

ACADEMIC REFERENCES

Prof. Dr. Michael Brand — Principal Investigator, CRTD, TU Dresden, Germany
Email: michael.brand@tu-dresden.de | Master's Thesis PI

Paul Röppenack — Research Supervisor, CRTD, TU Dresden, Germany
Email: paul.roeppenack@tu-dresden.de | Master's Thesis Supervisor